INTRO TO git + GitHub

Session by
Shorook Saleh

What is Version Control?

The magic of controlling history!

Version Control System (VSC)

A version control system, or VCS, tracks the history of changes as people and teams collaborate on projects together.

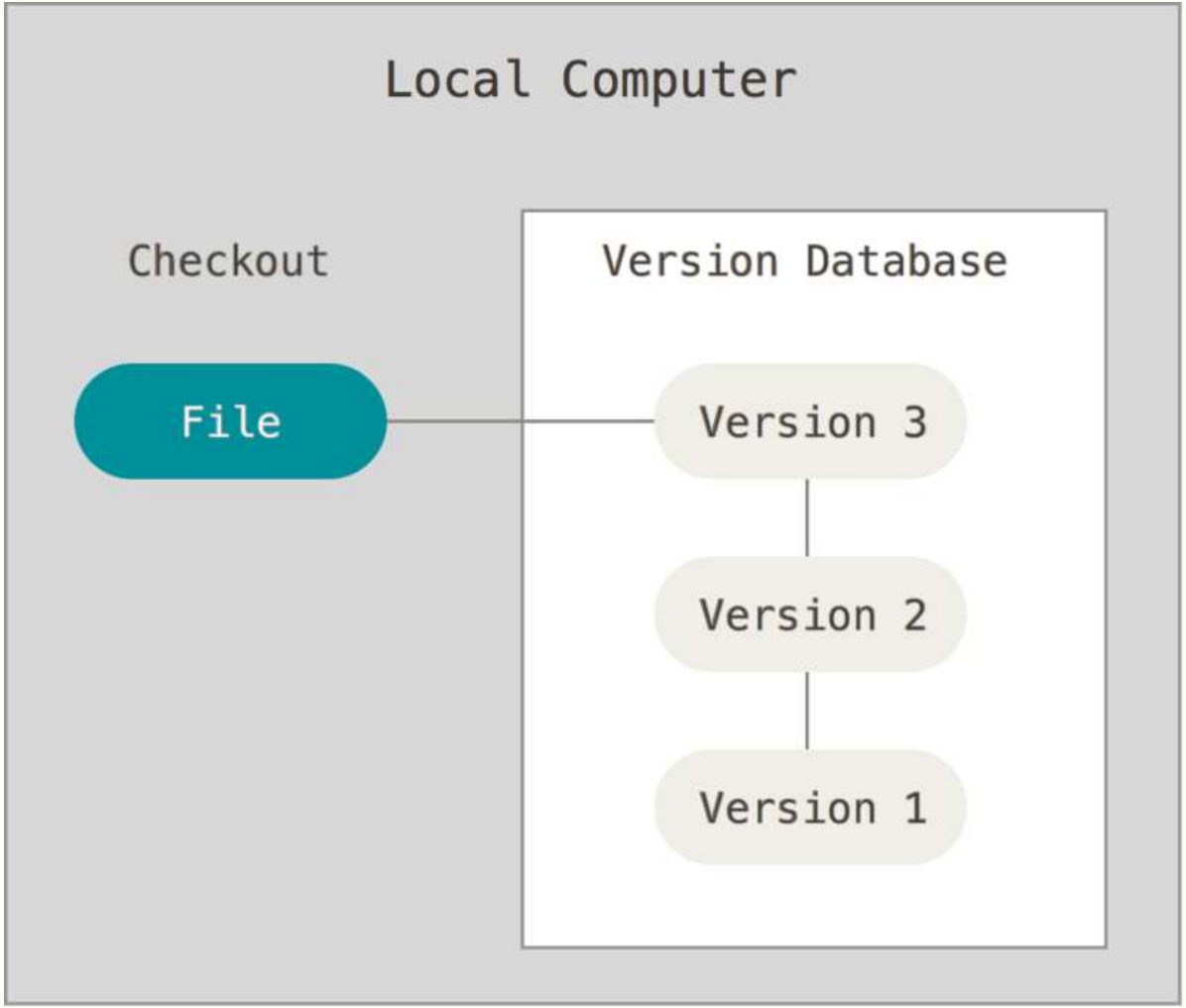
Developers can review project history to find out:

- Which changes were made?
- Who made the changes?
- When were the changes made?
- Why were changes needed?

Local Vs Distributed Vs Centralized

Local VCS

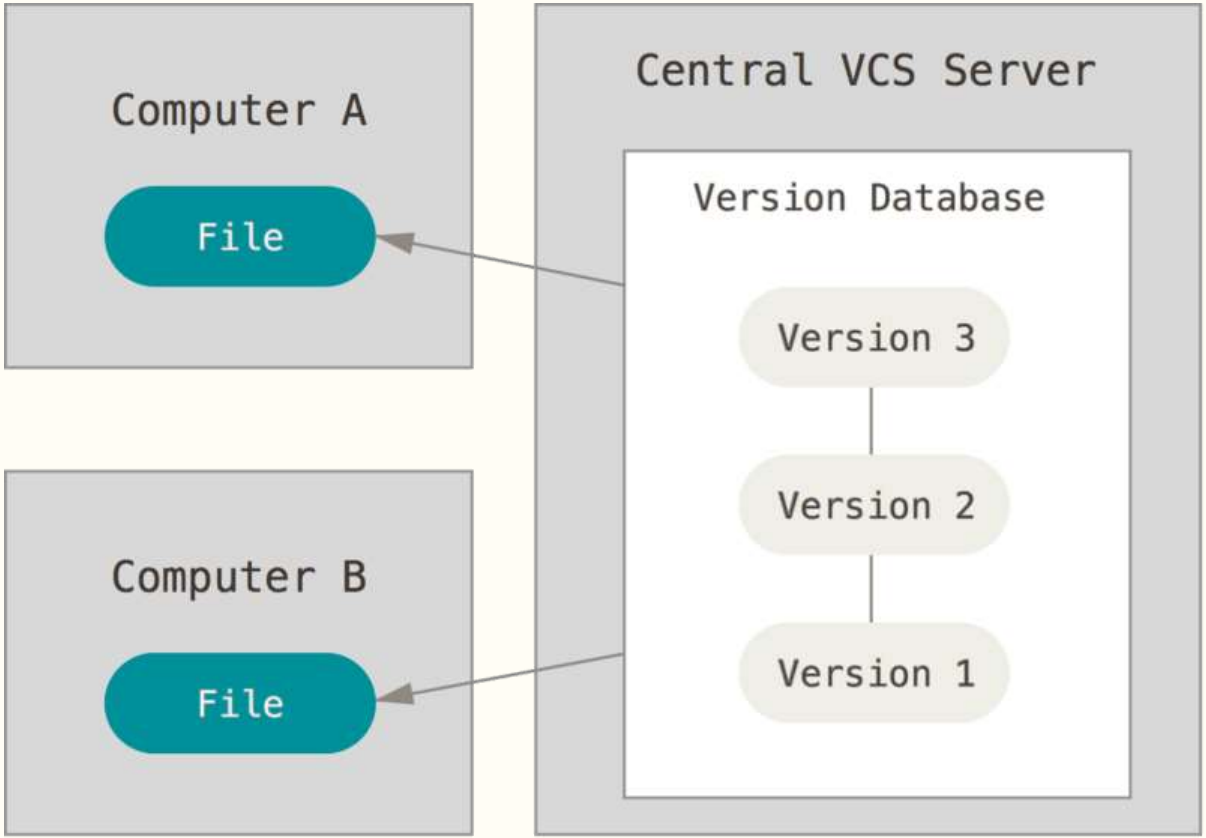
RCS works by keeping patch sets (that is, the differences between files) in a special format on disk; it can then re-create what any file looked like at any point in time by adding up all the patches.



Centralized VCS

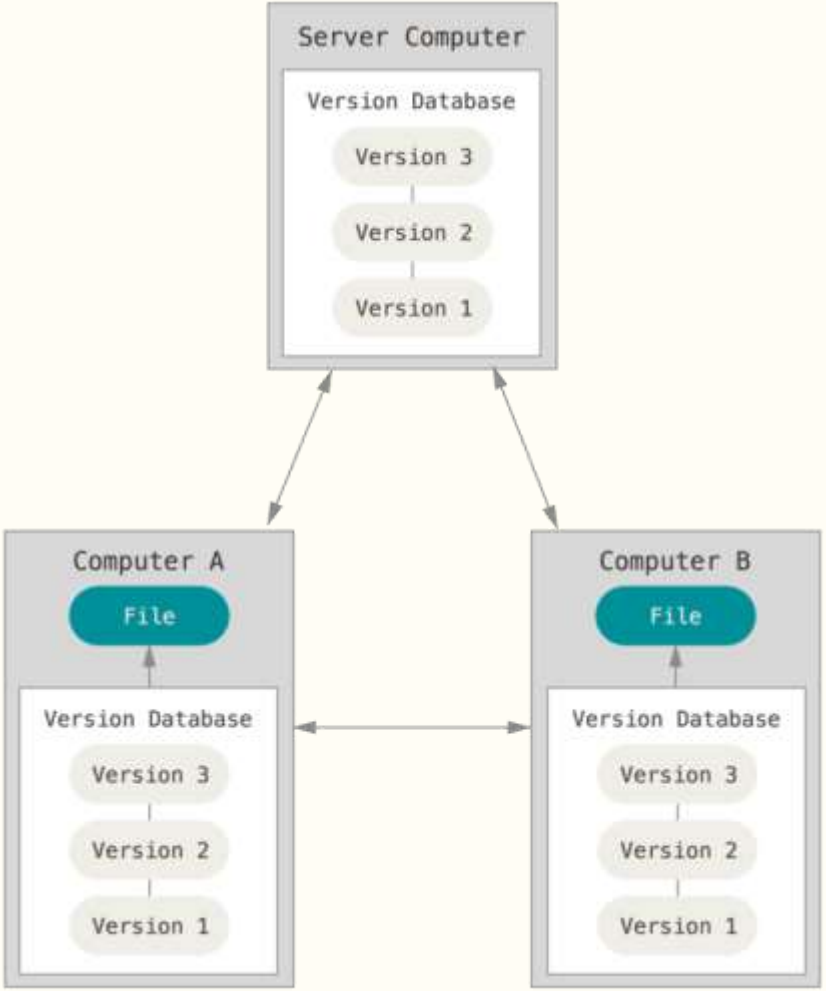
These systems (such as CVS, Subversion, and Perforce) have a single server that contains all the versioned files, and a number of clients that check out files from that central place. For many years, this has been the standard for version control.

However, this setup also has some serious downsides. The most obvious is the single point of failure that the centralized server represents.



Distributed VCS

In a DVCS (such as Git, Mercurial, Bazaar or Darcs), clients don't just check out the latest snapshot of the files; rather, they fully mirror the repository, including its full history. Thus, if any server dies, and these systems were collaborating via that server, any of the client repositories can be copied back up to the server to restore it.



What is Git?

I don’t git it?!

A Short History on Git!

In 2005, the relationship between the community that developed the Linux kernel and the commercial company that developed BitKeeper broke down, and the tool's free-of-charge status was revoked. This prompted the Linux development community (and in particular Linus Torvalds, the creator of Linux) to develop their own tool based on some of the lessons they learned while using BitKeeper.

Goals of the new system:

- Speed
- Simple Design
- Fully distributed
- Able to handle large projects
- Strong support for non-linear development

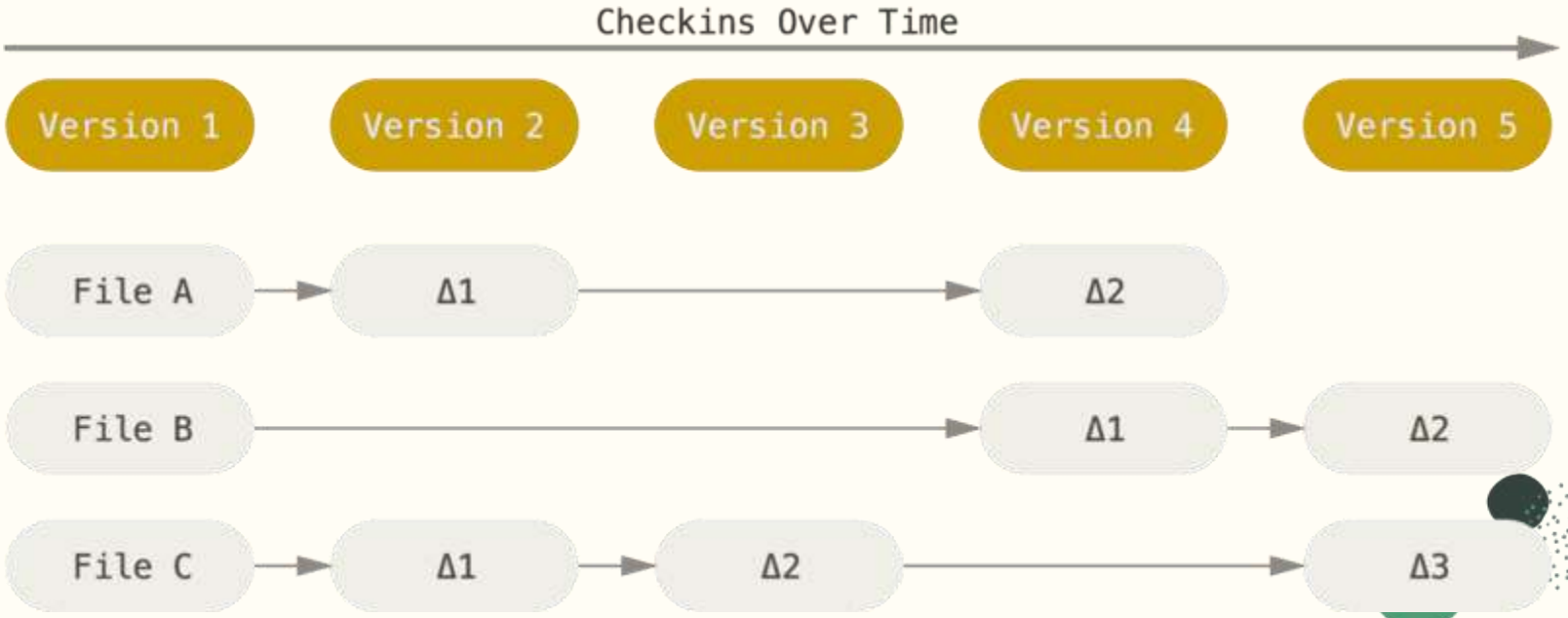
What is Git?

<u>The main differences that defines git from the other VCS :</u>

- Snapshots, Not Differences
- Nearly Every Operation Is Local
- Git Has Integrity
- Git Generally Only Adds Data
- The Three States

What is Git?

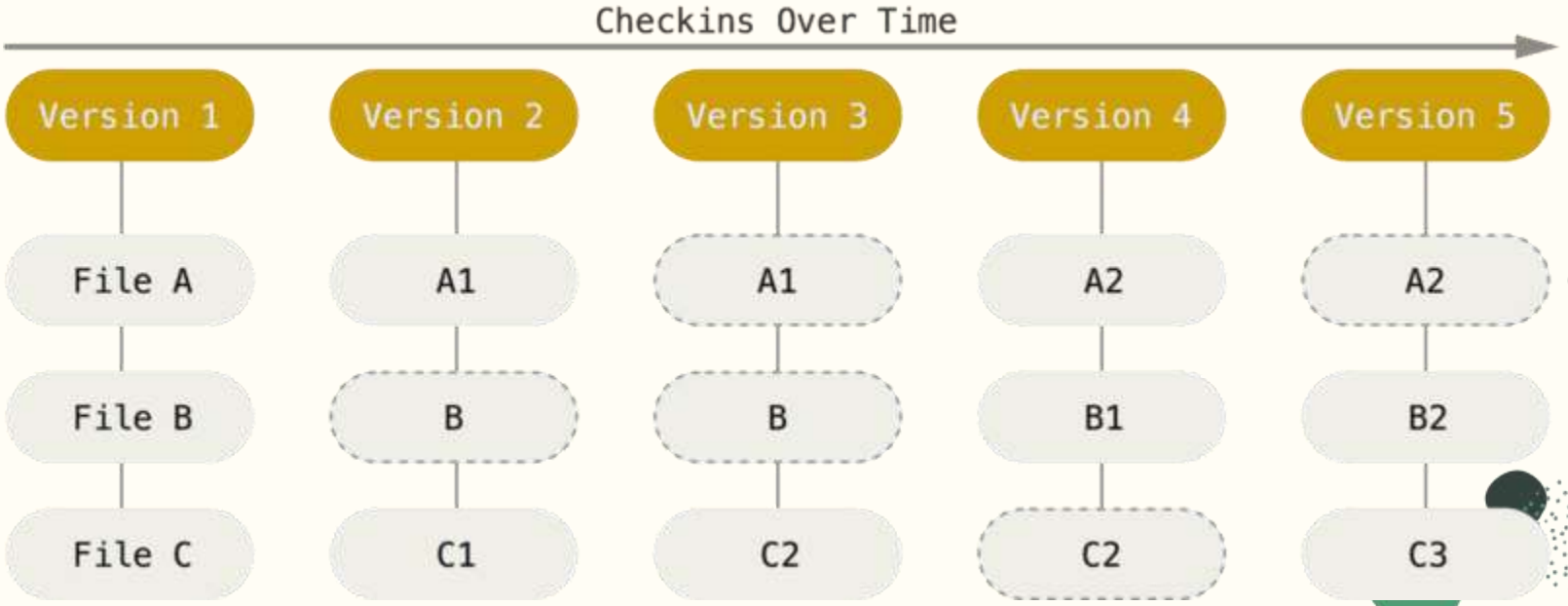
Snapshots, Not Differences



delta-based version control

What is Git?

Snapshots, Not Differences



Git’s stream of snapshots.

What is Git?

Nearly Every Operation Is Local

Most operations in Git need **only local files** and resources to operate.

- No need to connect to an external server so your code could work
- You can browse the entire history of your repo from your machine
- You can do almost anything offline like edit and commit changes locally

Git Has Integrity

Everything in Git is checksummed using SHA-1 before it is stored and is then referred to by that checksum.

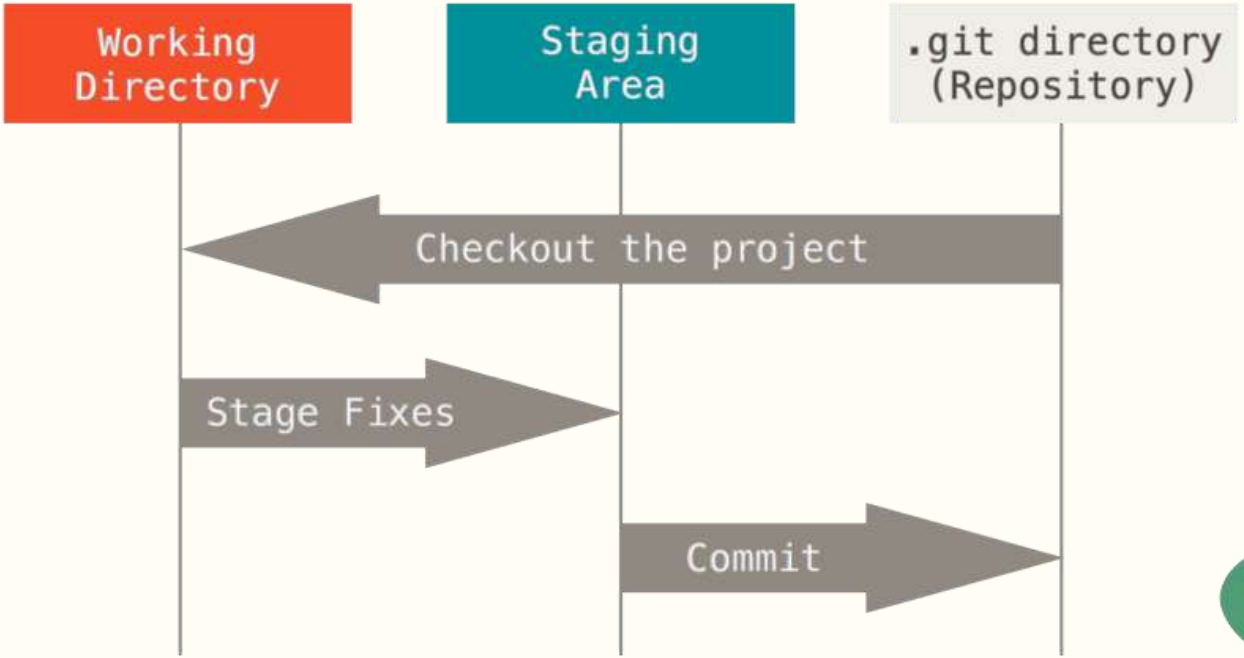
24b9da6552252987aa493b52f8696cd6d3b00373

Git Generally Only Adds Data

In Git you can't screw things up, because all actions are nearly "Add. After you commit a snapshot it is very difficult to lose.

What is Git?

The Three States



Git has three main states that your files can reside in: **modified**, **staged**, and **committed**

Git Commands

Git Basics

Initialize repo	`git init`
Add files to staging	`git add <files>`
Commit files	`git commit -m 'initial project version'`
Clone repo	`git clone <url>`
Push changes to upstream	`git push`
Pulling the code from remote repo	`git pull`
Setting up the upstream origin to the current master	`git push -u origin master`
Shows the status of changes (untracked, modified, staged)	`git status`
Shows the branches locally	`git branch`

What is Github?

They have an Octocat...

GitHub

What GitHub is?

GitHub is a Git hosting repository that provides developers with tools to ship better code through command line features, issues (threaded discussions), pull requests, code review, or the use of a collection of free and for-purchase apps in the GitHub Marketplace.

How Github works?

GitHub builds collaboration directly into the development process. Work is organized into repositories, where developers can outline requirements or direction and set expectations for team members.

GitHub Flow

The GitHub flow is a lightweight, branch-based workflow built around core Git commands used by teams around the globe—including ours.

GitHub flow's six steps:

GitHub Hello World

Create your first repository

Follow instructions on link here

https://guides.github.com/activities/hello-world/

Installing Git

Installing

Linux

- Fedora using yum `$ sudo yum install git`
- Debian-based ex Ubuntu `$ sudo apt-get install git`

Mac OS

Git comes preinstalled for macOS and you can find out using `$ /usr/bin/git --version`

Otherwise you use Homebrew to install it or install Xcode Command Tools.
On (10.9) and above you can simply do that by trying to run Git from the Terminal the very first time.

Windows

The Official build is on http://git-scm.com/download/win

A cool article “5 Ways to download git for windows”
https://www.jamessturtevant.com/posts/5-Ways-to-install-git-on-Windows/

Thank You!

GDG Kuwait

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Shorook Saleh

https://github.com/shorookMS

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